Lab#07

Construct if else statements in java

Objective

Implementation of various decisions making constructs.

Lab Goals

To provide working insight into the usage of decision making statements using the following:

a. If statement.

b. If then else statement.

c. Nested If statement.

Introduction to Decision making statements

Decision making statements contain conditions that are evaluated by the program. If the condition is true, then a set of statements are executed and if the condition is false then another set of statements is executed. ... Otherwise, statements in the body of else are executed.

If statement

If statement is the simplest decision making statement. It is used to decide whether a certain statement or block of statements will be executed or no

Syntax

```
if (condition) {
  // Statements to execute if
  // condition is true
}
```

Example#1

```
public class Main {
    public static void main(String[] args) {
        int x = 10;
        if (x > 5) {   System.out.println("x is greater than 5");   }
    }
}
```

Output

```
x is greater than 5
```

Exercise

A. Take values of length and breadth of a rectangle from user and check if it is square or not.

Program:

```java
import java.util.Scanner;

public class Lab7 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter length  : ");
        int length = sc.nextInt();

        System.out.print(s: "Enter breadth : ");
        int breadth = sc.nextInt();

        if (length == breadth) {
            System.out.println(x: "It is a Square!");
        } else {
            System.out.println(x: "It is NOT a Square.");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 7> javac Lab7.java
PS C:\Users\Ansharah\Desktop\Java\Lab 7> java Lab7
Enter length  : 15
Enter breadth : 56
It is NOT a Square.
PS C:\Users\Ansharah\Desktop\Java\Lab 7>
```

B. A school has following rules for grading system:

a.	Below 25	F
b.	25 to 45	E
c.	45 to 50	D
d.	50 to 60	C
e.	60 to 80	B
f.	Above 80	A

Write a program to enter marks and print the corresponding grade.

Program:

```java
import java.util.Scanner;

public class Lab7 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter marks: ");
        int marks = sc.nextInt();

        if (marks < 25) {
            System.out.println(x: "Grade: F");
        } else if (marks < 45) {
            System.out.println(x: "Grade: E");
        } else if (marks < 50) {
            System.out.println(x: "Grade: D");
        } else if (marks < 60) {
            System.out.println(x: "Grade: C");
        } else if (marks <= 80) {
            System.out.println(x: "Grade: B");
        } else {
            System.out.println(x: "Grade: A");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 7> javac Lab7.java
PS C:\Users\Ansharah\Desktop\Java\Lab 7> java Lab7
Enter marks: 89
Grade: A
PS C:\Users\Ansharah\Desktop\Java\Lab 7>
```

C. A student will not be allowed to sit in exam if his/her attendance is less than 75%. Take following input from user, Number of classes held, Number of classes attended and print percentage of class attended. Is student is allowed to sit in exam or not.

Program:

```java
import java.util.Scanner;

public class Lab7 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter total classes held    : ");
        int total = sc.nextInt();

        System.out.print(s: "Enter classes attended      : ");
        int attended = sc.nextInt();

        // Calculate attendance percentage
        double percentage = (attended * 100.0) / total;

        System.out.println("Attendance Percentage       : " + percentage + "%");

        if (percentage >= 75) {
            System.out.println(x: "Student IS allowed to sit in exam.");
        } else {
            System.out.println(x: "Student is NOT allowed to sit in exam.");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 7> javac Lab7.java
PS C:\Users\Ansharah\Desktop\Java\Lab 7> java Lab7
Enter total classes held     : 15
Enter classes attended       : 14
Attendance Percentage        : 93.33333333333333%
Student IS allowed to sit in exam.
PS C:\Users\Ansharah\Desktop\Java\Lab 7>
```

D. **Create a console-based Registration Form using Java.**

Your program should ask the user to enter the following details:

1. Full Name
2. Age
3. Email
4. Password

Perform **validation using an if-else if-else ladder**:

- Full Name: Should not be empty.
- Age: Must be between 18 and 60.
- Email: Must contain `@` and a `.` after `@`.
- Password: Must be at least 6 characters long.

Display a message indicating whether registration is **successful** or what the validation error is.

Program:

```java
import java.util.Scanner;
public class Lab7{
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Take user input
        System.out.print(s: "Enter Full Name : ");
        String name = sc.nextLine();

        System.out.print(s: "Enter Age       : ");
        int age = sc.nextInt();
        sc.nextLine(); // clear buffer

        System.out.print(s: "Enter Email     : ");
        String email = sc.nextLine();

        System.out.print(s: "Enter Password  : ");
        String password = sc.nextLine();

        // Validation using if-else ladder
        if (name.isEmpty()) {
            System.out.println(x: "Error: Full Name cannot be empty.");
        } else if (age < 18 || age > 60) {
            System.out.println(x: "Error: Age must be between 18 and 60.");
        } else if (!email.contains(s: "@") || !email.contains(s: ".")) {
            System.out.println(x: "Error: Email must contain @ and a dot (.)");
        } else if (password.length() < 6) {
            System.out.println(x: "Error: Password must be at least 6 characters.");
        } else {
            System.out.println("Registration Successful! Welcome, " + name);
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 7> javac Lab7.java
PS C:\Users\Ansharah\Desktop\Java\Lab 7> java Lab7
Enter Full Name : Ansharah Khan
Enter Age       : 19
Enter Email     : ansharahkhan81@gmail.com
Enter Password  : 2244667
Registration Successful! Welcome, Ansharah Khan
PS C:\Users\Ansharah\Desktop\Java\Lab 7>
```